

The foreign experience and investment performance of venture capitalists

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Abstract

This study examines the impact of venture capitalists with foreign experience on investment performance. Based on venture capitalist- and deal-level samples, venture capitalists with foreign experience are found to yield significantly worse investment performance than those with domestic experience in China. The study findings hold across a wide range of robustness checks. Further results indicate that venture capitalists with foreign experience are more prone to follow prudent investing strategies, invest shortly before exiting and conduct less supervision. One underlying mechanism through which venture capitalists with foreign experience yield poorer performances is cultural distance.

Key words: Cultural distance; Foreign experience; Investment performance; Venture capitalists

JEL classification: G24, G30

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1. Introduction

The human capital of venture capitalists is a fundamental factor that concerns successful investment within the venture capital (VC) industry, and it is, on average, two to five times more important than a VC firm's

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organisational capital in explaining investment performance (Kaplan and Schoar, 2005; Hochberg *et al.*, 2014; Ewens and Rhodes-Kropf, 2015). This is because venture capitalists with different forms of human capital – for example, venture capitalists who used to be government officials, commercial bankers, or entrepreneurs – bring potential benefits or costs to their portfolio companies and impact their investment performance (Hellmann and Puri, 2002; Wang and Wu, 2020). However, most previous studies focus on VC's capital background or organisational structure (Hochberg *et al.*, 2007; Hellmann *et al.*, 2008; Fulghieri and Sevilir, 2009; Butler and Goktan, 2013; Fang *et al.*, 2013; Humphery-Jenner and Suchard, 2013; Chemmanur *et al.*, 2014), with infrequent attention paid to the human capital effects embedded in VC firms. Unlike existing studies that explore venture capitalists' ethnic, educational (e.g., science and engineering, business administration master's degree, certificate related to law, and university ranking), and professional (e.g., entrepreneurial, industrial, and consultancy-related aspects) background (Dimov and Shepherd, 2005; Bottazzi *et al.*, 2008; Patzelt *et al.*, 2009; Rebecca, 2010; Gompers *et al.*, 2016), we closely focus on the human capital accumulated by venture capitalists through foreign experience, which features prominently in emerging markets.

Our study results indicate that venture capitalists with foreign experience yield significantly worse investment performances than those with domestic experience. Moreover, we show that these worse performances are accompanied by a series of prudent investing strategies. First, venture capitalists with foreign experience prefer multistage and syndicate investments. Second, they are less ambitious concerning leading investors. Additionally, venture capitalists with foreign experience have low board participation, thus indicating that their supervision of startup firms is inadequate. Finally, we identify that the mechanism through which the venture capitalists with foreign experience yield worse performance is cultural distance.

China provides a unique and advantageous setting to explore whether foreign experience is effective in the VC market. China's VC industry was established in the mid-1980s, when China decided that it should develop its local high-tech industries (Pukthuanthong and Walker, 2007) while it had been expanding in developed markets over the past decades and had contributed to the rise of Silicon Valley in the United States. To compensate for the shortage of experience and expertise, the Chinese government and domestic VCs explored cooperation with foreign VC talent including establishing overseas offices.¹ Additionally, China's economy has a lasting attraction for foreign talent, relying on bright prospects and promising returns of the past few decades, and many well-known high-tech companies such as Baidu, Alibaba, and Tencent

¹This statement comes from *China Venture Capital Yearbook* (Cheng, 2012), China Venture Capital Research Institute, Shenzhen, China.

received support from foreign venture capitalists at their inception or startup development stages (Allen *et al.*, 2005; Que and Zhang, 2018). Chinese culture, which features a set of beliefs related to Confucius rather than to a religion, considerably differs from the cultures of many other countries (Allen *et al.*, 2005). Considering that national cultural distance represents the distance among the norms and routines of and the repertoires for, organisational design, new product development, and other management aspects (Dai and Nahata, 2016), it is difficult to ascertain whether foreign experience has a definitive impact in China.

Thus, the present study examines the impact of talented venture capitalists with foreign experience on investment performance in China. First, we file all investment events that occurred from 2001 to 2012 and the list of involved venture capitalists and screen out the *de facto* managers during investment implementation according to their position and related reports and take them as the subjects of follow-up analysis. We manually collect data pertaining to these venture capitalists' foreign education, working experience, and other demographic characteristics from their bios. Subsequently, the average returns and the proportions of deals that have been successfully exited through initial public offerings (IPOs) between venture capitalists with foreign experience and those with domestic experience are examined. We find that venture capitalists with foreign experience produce poorer returns and lower IPO rates. In particular, they produce 9.68 percent less internal rate of return (*IRR*) and 15.88 percent less return on investment (*ROI*) than those with domestic experience. Our baseline results are robust to deal-level data (capitalist-company-deal observations) and a subsample analysis of venture capitalists who are affiliated to domestic VC firms.

Although we endeavour to sample as many of the VC events for the aforementioned period as possible, it is potentially subject to survivorship and a sample selection problem because of data limitations: venture capitalists with foreign experience are usually well-educated and may be left out for language or business reasons. Conversely, some venture capitalists with domestic experience may choose not to disclose their personal information. To address this concern, we use the propensity score matching procedure, which allows us to minimise the difference in observable characteristics between the two types of venture capitalists. We find that venture capitalists with foreign experience consistently show poorer performance.

Further, we draw insight regarding the investing strategies of the two venture capitalist groups and find that venture capitalists with foreign experience tend to syndicate their investments with other venture capitalists, rather than invest alone; they also tend not to act as a leader (the one who disburses capital first). Additionally, they prefer investing with multistage methods. We also find that venture capitalists with foreign experience are less likely to be directors of the startup companies backed by them and that they carry out shorter investment

duration compared with those with domestic experience. Therefore, we conclude that venture capitalists with foreign experience are more likely to take prudent investing strategies and undertake less supervision for the entrepreneurial firms after investing.

We are interested in the mechanism that drives our main effect, in particular whether venture capitalists with foreign experience yield worse performance because of their inferior ability to nurture start-ups (a treatment effect) or to screen firms with high potential or a good price (a selection effect). To disentangle the two different effects, we employ a Heckman two-stage regression because the selection effect implies a potential endogeneity or selection bias issue in our result, that is, it asserts that the better performance of venture capitalists with domestic experience is driven by a selection bias in that they are merely picking companies of better quality. We estimate the selection model using the density of venture capitalists with foreign experience in the startup company's city as our exogenous instrument. Our results indicate that the baseline findings are consistent after correcting for this selection bias and venture capitalists with foreign experience still yield worse performance. Therefore, we disentangle the treatment effect from the selection effect to a certain degree.

Finally, we explore the nature of difference between the two venture capitalist groups. As mentioned, culture is particularly essential in business; therefore, we study the impact of cultural distance on the investing performance. We observe that venture capitalist investment performance decreases with the increase of cultural distance between the experience country and China. Additionally, companies backed by venture capitalists with foreign experience are more likely to take an IPO overseas, while companies backed by venture capitalists with domestic experience have a higher possibility to take a native IPO. This aspect is consistent with the imprinting theory that is proven to occur in CEOs (Bai *et al.*, 2020).

Further, our paper makes important contributions to several strands of the literature. First, our paper provides a deeper understanding of the drivers of performance persistence of VC funds. Prior to our analysis, there were several notable studies that shed light on this topic. Kaplan and Schoar (2005) suggest that the performance of VCs appears highly persistent across a sequence of funds managed by the same manager. A theoretical model and empirical evidence show that highly persistent fund returns are derived from a VC firm's skill factor (Hochberg *et al.*, 2014). Further, Ewens and Rhodes-Kropf (2015) state that skill and exit style differs even among venture partners in the same VC firms. Then, they conclude that partners' human capital is two to five times more important than the VC firm's organisational capital in explaining performance. Our study is therefore consistent with these discoveries and directly explores the impact of venture capitalists' overseas experience on the performance.

It also extends the existing empirical literature on cross-border venture capital, which has a large gathering of venture capitalists with foreign experience. The extant literature compare foreign and domestic VC performance from the standpoint of VC firms and procure similar results: foreign VCs are no better at investing performance and promoting companies to go public and are at a disadvantage concerning information collection and monitoring due to geographical and cultural distance (Wang and Wang, 2011, 2012; Dai *et al.*, 2012; Humphery-Jenner and Suchard, 2013; Li *et al.*, 2014; Chemmanur *et al.*, 2016), which is consistent with our results. Specifically, Wang and Wang (2011) highlight that little correlation exists between VC's performance and its foreign background, network or reputation. Dai *et al.*, (2012) and Li *et al.*, (2014) attribute a part of foreign VC's failing to cultural distance related to VC's headquarter location and investing country. Our paper complements these studies by examining the cultural difference that arises from venture capitalists' working and education experiences. Additionally, Wang and Wang (2012) reveals that the domestic country's economic freedom is crucial to cross-border VC performance, which inspires us to probe into venture capitalists' performance in emerging markets. Finally, some previous studies focuses on the strategy differences between VCs that invest in emerging and advanced economies (Bruton and Ahlstrom, 2003; Baum and Silverman, 2004). Unlike these studies, our paper illustrates the operating discrepancy of venture capitalists that come from different economies and invest in the same emerging area.

Third, our paper contributes to the literature exploring cultural disparities. Differences among national cultures have been considered to play a vital role in cross-border transactions (Morosini *et al.*, 1998; Brouthers and Brouthers, 2001; Dai and Nahata, 2016). International corporations select high- or shared-control entry modes to avoid cultural friction when they set up their foreign subsidiaries (Brouthers and Brouthers, 2001), and foreign VCs seek local VCs or locally experienced foreign VCs for syndication to implement monitoring strategies and obtain eventual success (Dai and Nahata, 2016). While much of the academic literature deems cultural disparity as an inherent difference between countries, cultural transmission has received scant attention. We thus analyse how personal education and work experience influence their cultural attributes and final economic outcomes.

The remainder of this paper is organised in the following. Section 2 provides a brief literature review and develops the hypotheses. Section 3 contains the data and sample descriptions. Section 4 presents the empirical results and robustness tests. Section 5 examines the mechanisms that cause inferior performance by venture capitalists with foreign experience. Finally, Section 6 concludes the paper.

2. Literature review and hypothesis

2.1. Foreign experience, cultural distance and investment performance

The literature offers two aspects of explanations regarding who are more successful – venture capitalists with foreign experience or domestic experiences. The first strand of literature champions the view that venture capitalists with foreign experience yield better investment performance in the emerging markets through at least three channels. First, having learned how foreign organisations function, venture capitalists with foreign experience can facilitate the adoption of superior management practices that have been shown to enhance the efficiency of investment and productivity that plays a decisive role in a deal's outcome (Bloom and Van Reenen, 2007; Dai and Liu, 2009; Li *et al.*, 2012). The empirical evidence shows that performance increases after firms hire directors with foreign experience and identify the channels through which the emigration of talents may lead to a brain gain (Giannetti *et al.*, 2015).

Second, venture capitalists with foreign experience may have connections in foreign countries that facilitate foreign acquisitions and international capital-raising activities (Thapa *et al.*, 2013). Further, strong relationship with overseas markets may indirectly provide access to international service providers such as headhunters and prestigious investment banks (Dai *et al.*, 2012).

Third, venture capitalists can serve as a catalyst to new venture internationalisation when these capitalists have adequate international knowledge and reputation (Fernhaber and McDougall-Covin, 2009). For example, venture capitalists with abundant information and strong reputation worldwide likely improve corporate innovation, customer expansion, and multinational coalition possibilities for their portfolio companies (Mäkelä and Maula, 2005; Fernhaber and McDougall-Covin, 2009; Filatotchev *et al.*, 2009; Luong *et al.*, 2017; Yuan and Wen, 2018).

By contrast, the other strand of literature favours venture capitalists with domestic experience and claims that such capitalists face fewer investment barriers than their counterparts for culture convergence. First, venture capitalists that have studied or worked in the host country are more likely to have behaviours and concepts that are in line with the portfolio company's culture. The cultural disparities between the investors and their companies can manifest in the level of confidence in each other, nature of financial contracting, and potentially severe conflicts (Pruthi *et al.*, 2003; Dai and Nahata, 2016). For example, in a country such as China, which exhibits a strong collectivistic culture, employees tend to share responsibility within an organisation (Boisot and Child, 1988; Pukthuanthong and Walker, 2007). This means that negotiations and routine communications will rarely occur on a one-on-one basis and will typically entail communicating with several individuals who share responsibility for a given task. This collectivistic approach is far reaching for Western business people (Ueno and Sekaran, 1992).

Second, the business and production of the majority of start-ups in emerging markets still focus on domestic markets, although the movement of capital and talent is transnational (Strother, 2014). The skills and experience that venture capitalists gained abroad while studying and working are more likely to be suitable for overseas cultures. Cultural disparity is an important reason for the failure of product standardisation and decrease in foreign capital inflow (Douglas and Wind, 1987; Du *et al.*, 2012). The exit of international giants such as Amazon and Best Buy from the Chinese market is related to the different in habits of Chinese consumers, although these companies have advanced management, overseas financing, and international markets (Zhao and Jin, 2016).

Finally, venture capitalists with domestic experience have a deeper understanding of laws and regulations, which are characterised by local culture; such an understanding facilitates the supervision and guidance for start-ups. Compared to the Western market where business transactions are typically based on carefully worded contracts that are enforceable under a country's laws and regulations, emerging markets have relatively more flexible policies (Li *et al.*, 2008). For example, firms need to apply for a licence before starting a new business in China (Allen *et al.*, 2005; Wang and Wu, 2020), and issuing an IPO requires the issuer to fulfil the requirements of the Chinese Securities and Regulatory Commission (Humphery-Jenner and Suchard, 2013). Additionally, Chinese business partners rely more heavily on the social responsibilities that come with a set of traditional beliefs, which is first developed and formalised by Kong Zi (Confucius) (Pye, 1982; Chow, 2002).

Whether a venture capitalist's foreign experiences are associated with better investment performance is therefore an intriguing and ultimately empirical question. Thus, we explore this issue by testing the following hypothesis; the alternate hypothesis predicts exactly the opposite:

H1: *Ceteris paribus*, venture capitalists with foreign experience yield a worse performance compared to those with domestic experience

If cultural difference is an important reason for the performance discrepancy between the two venture capitalist groups, the investment performance of venture capitalists with foreign experience should be different with cultural distances between experienced countries and China. Therefore, our second hypothesis is given below:

H2: *Ceteris paribus*, the investment performance of venture capitalists decreases with an increase in the cultural distance between the experienced countries and China

2.2. Foreign experience and investment strategies

National cultural distance raises barriers to information sharing, reduces trust, and increases transaction costs (Li *et al.*, 2014), which play an important role in shaping the investors' risk attitude toward investments and can ultimately affect investment decisions (Kalev *et al.*, 2008). Investors with an advantage tend to act

alone to prevent others from hitchhiking (Cestone *et al.*, 2007; Bottazzi *et al.*, 2016). Therefore, venture capitalists with domestic experience may present seemingly ‘aggressive’ and exclusive strategies. Conversely, venture capitalists with foreign experience present ‘prudent’ and cooperative strategies. This aspect is consistent with Tian (2011) and Dai and Nahata (2016), who indicated that local venture capitalists reduce investment staging by increasing the average round amount compared to foreign venture capitalists.

On the other hand, venture capitalists implement supervision and advising activities for management through membership on the board of directors (Fried *et al.*, 1998; Hellmann and Puri, 2002), and prior empirical tests have shown that board membership is an effective way to increase investment returns (Hellmann and Puri, 2002). Nevertheless, cultural disparity is an important hidden danger in enterprise management and might trigger severe conflicts, as previously analysed.

Based on the aforementioned analysis, venture capitalists with foreign experience are likely to implement prudent investment strategies beforehand instead of strict management during the process of reducing risk while avoiding friction with the management. Therefore, our third hypothesis, based on the preceding discussion, is stated below:

H3: *Ceteris paribus*, venture capitalists with foreign experience perform more ‘prudent’ investment strategies but less supervision for their entrepreneurial firms

3. Materials and methods

3.1. Data sources and sample construction

The main data set begins with VC financings, institutions, venture capitalists, and entrepreneurial firms from the CVSource database. Using regular surveys, press releases, and regulatory filings, CVSource comprehensively portrays the Chinese VC market. The full database covers 1990–2017 and includes 34,338 investment events (excluding post-IPO equity investments and investments in films or television series). Our sample only includes investment events that occurred between 2001 and 2012 (a period during which the amount of domestic and foreign VC is relatively balanced) and the involved venture capitalists, which allows private firms at least five years to exit. For every deal, we screen out the *de facto* managers during investment implementation according to their positions and related reports, that is, the then-head of the VC firm who dominated and played a decisive role in the transaction as the venture capitalist. Generally, the head of the firm is listed at the beginning and titled as ‘Founding Partner’, ‘Principal’, or ‘Managing Director’, and so on. Following previous studies, we do not include investment events for financial corporations.

Finally, we collected 10,253 investment events, 4,671 entrepreneurial firms, 859 investing institutions, and 2,133 unique venture capitalists. Additionally,

we supplement the data on two fronts. First, we extend public companies' information from the China Stock Market & Accounting Research (CSMAR) and Wind databases, which are both leading data vendors; CSMAR can be obtained from Wharton Research. Second, we acquire missing information regarding private companies' detailed data and venture capitalists' bios through web searches. We attempt to preserve the integrity and consistency of the data, but some variables remain missing.

3.2. *Identifying venture capitalists with foreign experience*

We define venture capitalists as having foreign experience if they have studied at university or worked for institutions that are outside China; if not, they are defined as venture capitalists with domestic experience. We begin with a list of the 2,133 venture capitalists that have investing records between 2001 and 2012, as reported by the CVSource database. Using their bio information, we manually identify venture capitalists' educational and professional experience prior to having entered the VC industry. To ensure that foreign experience captures actual exposure to a foreign environment, we do not consider those working for a Chinese branch of a foreign company or a joint venture as having foreign work experience.

This definition implies that there are two types of venture capitalists that are classified as those with foreign experience: foreign capitalists who were born and grew up overseas before entering VC markets and Chinese capitalists who grew up in domestic areas but have overseas experience. Notably, we select university education and prior working experience as our screen standards because university education and working platforms are critical to human capital accumulation compared to other elements during one's lifetime.

3.3. *Measuring investment performance*

We employ several dependent variables of interest throughout the analysis. Initially, we use investment returns to capture investing performance, including the *IRR* and the *ROI*. These variables are better proxies because they capture the actual investment output and how effectively a venture capitalist has utilised the money and human capital. We obtain detailed information on investment returns from the CVSource database. The database provides the invested amount and total return amount of every completed investment.

Using return data to measure investing performance causes us to focus only on the exited deals. In other words, we do not involve ongoing deals in our analysis. Following the extant literature (Gompers and Lerner, 2000; Hochberg *et al.*, 2007; Nahata, 2008), we characterise success investment by observing whether the entrepreneurial firm has an IPO. In other words, we consider success by considering whether there is an opportunity for capital to exit through an IPO.

3.4. Control variables

To control the influence of other factors on investment performance, we construct control variables, including the gender of the venture capitalist (*Gender*), their education (*Edu_Master*, *Edu_PhD*), past investment quantity (*Experience*), fund source (*VC_Mainland*), enterprise's survival time at the investing year (*Comp_Age*), firm's location (*City*), and firm's industry (*Industry*). To mitigate the impact of investment strategy on investment performance, we also include the years from investing to exiting ($\ln(\textit{Duration})$) in our baseline regressions. Beyond the aforementioned, we have not found any other strategies with a significant impact on investment outcomes. Appendix Table A1 defines the variables used herein. Following the literature on VC, we obtain other venture capitalist characteristics from the bios and enterprise data through public channels. However, we could not control more characteristics of invested objects as majority of the start-up firms are private when they are invested and have no obligation to disclose their financial and other information to the public.

3.5. Descriptive statistics

The analysis is processed on two levels: the venture capitalist and investment deal level. In the first level, we compare and analyse individual investment performance directly. The observational unit in this level is the venture capitalists. The investment deal level is decided by a unique venture capitalist, an enterprise, and an investment date. Some proxy indicators, such as *IRR*, *ROI*, *IPO*, and *Duration*, are directly related to investment deals; therefore, we construct a series of personal indexes for venture capitalist-level analysis. For example, we use $\ln(\textit{IRR_Ave})$ as a venture capitalist's personal performance. The *IRR_Ave* is a weighted average *IRR*, which is weighted by all deal amounts invested by the venture capitalist from 2001 to 2012. $\ln(\textit{IRR_Ave})$ is the natural logarithm of *IRR_Ave*. Similarly, we construct $\ln(\textit{ROI_Ave})$ and $\ln(\textit{Duration_Ave})$. To avoid losing venture capitalist and deal observations with zero return or zero IPO rates, we add one to the actual values and then take the natural logarithm. Additionally, to control for potential factors that might influence the dependent variables, we calculate the average portfolio firms' age (*Comp_Age*); fraction of portfolio firms located in popular city (*City*); and fraction of portfolio firms belonging to the technology, media, and telecom (TMT) industry (*Industry*) when we conduct the venture capitalist-level analysis.

In Table 1, Panel A reports the summary statistics for the investment performances of venture capitalists with investing experience from 2001 to 2012. On average, a venture capitalist gains an *IRR* of 50.28 percent or an *ROI* of 70.84 percent. If the sample is broken down into two venture capitalist groups, we find that venture capitalists with foreign experience have a lower *IRR* and *ROI*, that is, an average of 48.87 percent *IRR* and 62.51 percent *ROI*

Table 1
Summary statistics

	N	Mean	SD	Max	Min
Panel A: venture capitalists' investing performance (observation unit: venture capitalist)					
<i>IRR_Ave: Full sample</i>	948	0.5028	0.5234	3.2381	−0.3185
<i>IRR_Ave: with foreign experience</i>	427	0.4887	0.5797	3.2381	−0.1278
<i>IRR_Ave: with domestic experience</i>	521	0.5144	0.4726	2.8360	−0.3185
<i>ROI_Ave: Full sample</i>	947	0.7084	0.6129	2.7699	−0.1452
<i>ROI_Ave: with foreign experience</i>	432	0.6251	0.6210	2.6562	−0.1147
<i>ROI_Ave: with domestic experience</i>	515	0.7782	0.5977	2.7699	−0.1452
<i>Duration_Ave: Full sample</i>	974	3.3375	1.9180	15.2219	0.0000
<i>Duration_Ave: with foreign experience</i>	443	3.0309	1.7107	12.8466	0.0000
<i>Duration_Ave: with domestic experience</i>	531	3.5934	2.0418	15.2219	0.0000
<i>IPOrate: Full sample</i>	2133	0.3233	0.3978	1.0000	0.0000
<i>IPOrate: with foreign experience</i>	986	0.2856	0.3577	1.0000	0.0000
<i>IPOrate: with domestic experience</i>	1147	0.3557	0.4267	1.0000	0.0000
Panel B: other variables (observation unit: venture capitalist)					
<i>Overseas</i>	2133	0.4623	0.4987	1.0000	0.0000
<i>Gender</i>	2121	0.8883	0.3151	1.0000	0.0000
<i>Edu_Master</i>	1568	0.6594	0.4740	1.0000	0.0000
<i>Edu_PhD</i>	1568	0.1550	0.3620	1.0000	0.0000
<i>Experience</i>	2133	5.3179	11.8327	255.0000	1.0000
<i>VC_Mainland</i>	2018	0.4321	0.4955	1.0000	0.0000
Panel C: other variables (observation unit: investment deal/capitalist-firm-date)					
<i>Multistage</i>	10253	0.3159	0.4649	1.0000	0.0000
<i>Syndicate</i>	10040	0.4949	0.5000	1.0000	0.0000
<i>Leader</i>	9421	0.7819	0.4130	1.0000	0.0000
<i>Director</i>	1372	0.4905	0.5001	1.0000	0.0000
<i>CD_4</i>	5506	19.0697	4.5115	23.3500	3.9600
<i>CD_6</i>	5487	16.9185	3.9565	19.5800	5.2600
<i>Comp_Age</i>	7438	80.1103	74.2160	1,261.7670	0.0333
<i>City</i>	9958	0.7782	0.4155	1.0000	0.0000
<i>Industry</i>	10253	0.5051	0.5000	1.0000	0.0000

This table reports the characteristics of the sample of venture capitalists and their entrepreneurial investments from 2001 to 2012. Panel A presents the summary statistics for venture capitalists' performance and duration, whereas Panel B presents the summary statistics for other control variables. The observation unit in Panels A and B are both venture capitalists. Panel C presents the summary statistics for specific investment deals. The unit of observation is deal matched by unique capitalist, entrepreneurial firm, and investing date. See Appendix Table A1 for variable definitions.

per year, whereas an average venture capitalist with domestic experience gains 51.44 percent *IRR* and 77.82 percent *ROI*.² All venture capitalists with and

²We minimise the influence of extreme errors of *IRR*, *ROI*, and *Duration* by winsorising the top and bottom 1% of observations; we present the results without winsorising to show that this step does not affect our conclusions.

without exit records are covered in the sample if the IPO rate – compared to *IRR* and *ROI* – is taken as the performance proxy. On average, 32.33 percent of entrepreneurial firms are backed by a venture capitalist that go public successfully. Moreover, venture capitalists with foreign experience yield a lower enterprise IPO rate (28.56 percent) than those with domestic experience (35.57 percent).

Panel B provides the summary statistics of the venture capitalists' characteristics. As shown, 46.23 percent of the venture capitalists have an overseas experience, and 88.83 percent of them are male. The proportion of venture capitalists with a master's degree is 65.94 percent; the proportion of venture capitalists with a Ph.D. is 15.50 percent. Additionally, each venture capitalist is in charge for five deals on average, and 43.21 percent of them invest with capital that comes entirely from domestic funds.

Panel C summarises the venture capitalist-firm-date level variables, and the observational unit is 'deal'. In the full sample, 31.59 percent of deals proceeded in multiple stages, 49.49 percent are conducted by syndication, 78.19 percent are charged by leading or unique investors, and 49.05 percent set the venture capitalist a seat on the firm's board. The average age of entrepreneurial firms is 80.11 months when the deal occurs. Finally, 77.82 percent of the deals proceed for entrepreneurial firms located in Beijing, Guangdong, Shanghai, Zhejiang, and Jiangsu, and 50.51 percent of deals are an investment for the TMT industry.

4. Empirical results

The present study discusses the impact of venture capitalists' prior working or education location on investment performance. In our baseline analysis, we investigate the investment performance of two venture capitalist groups in subsection 4.1. Further, we conduct the baseline analysis again with a deal-level sample and verify our results in subsection 4.2. Subsequently, we examine venture capitalists' performance using propensity score-matched pairs of the deal to avoid survivorship bias in subsection 4.3. In subsection 4.4, we analyse the difference between the effects of foreign working and education experiences on investment performance. Finally, we compare the investment strategies of venture capitalists with different experiences in subsection 4.5.

4.1. Baseline findings

We begin by specifying our baseline regression. Specifically, to test Hypothesis 1, we use various forms of the following model to examine the impact of foreign experience on investment performance with a venture capitalist-level sample:

$$Performance = \alpha + \beta \cdot Overseas + \gamma \cdot Duration + \delta \cdot X + \varepsilon, \quad (1)$$

where the dependent variables are $Ln(IRR_Ave)$, $Ln(ROI_Ave)$, and $Ln(IPOrate)$ in different regressions. The key explanatory variable in the above model is *Overseas*, an indicator variable for whether a venture capitalist has foreign experience. *Duration* is, as mentioned earlier, an important control variable. X represents a vector of venture capitalists, funds, and company characteristics that could affect an investment's performance, as discussed in subsection 3.4. Meanwhile, *Stage* captures a start-up's development phase. All these variables are averaged at the venture capitalist level.

Table 2 reports the regression results for the investment returns of two venture capitalist groups, with *IRR*, *ROI*, and *Duration* winsorised to limit the influence of outliers (results using non-winsorised data are provided in Appendix Table A2 and show similar patterns). As per columns (1) and (2), where the dependent variable is the average *IRR* of investments filed by every venture capitalist, the coefficient estimate of *Overseas* is negative and statistically significant, suggesting that venture capitalists with foreign experience are associated with a lower inner rate of return. Economically, based on the coefficient estimate of *Overseas* dummy in Column (2), a venture capitalist with foreign experience generates a 9.68 percent lower *IRR* than those with domestic experience. Columns (3) and (4) present a similar analysis for *ROI*. The coefficient estimate of *Overseas* is negative and significant, suggesting that venture capitalists with foreign experience receive 15.88 percent lower *ROI*.

Next, we examine whether venture capitalists' foreign experience affects their investment's IPO rate, given that testing only the returns may result in a sample bias. Specifically, returns for an individual deal are unsystematically available because VC institutions disclose their performance only to their investors, and outstanding or attractive returns are more liable to be captured by media and databases. Compared to investment returns, the IPO status of entrepreneurial firms is more public. Moreover, IPO rates are highly correlated with investment returns. The venture capitalist's 'IPO rate' was defined as the fraction of deals that are successfully exited via an IPO.

Venture capitalists with foreign experience are found to suffer significantly inferior IPO-exit consequences (Table 3). After controlling for investment duration, venture capitalist characteristics (gender, education, and experience), fund source, industry, entrepreneurial firm's location, and investing round, the R-squared increases to approximately 24 percent. The economic magnitude of these effects is meaningful. As reported in columns (3) and (4), deals taken by venture capitalists with foreign experience receive 4.87–5.37 percent less IPO-exits than those with domestic experience. Note that in columns (2) and (4), the results remain consistent as before, although the investment duration (*Duration*) is controlled for and venture capitalists without exit records are excluded from the sample.

Subsequently, we re-estimate our venture capitalist-level model with venture capitalists who worked for institutions that gather all capital from domestic markets rather than overseas or mixed markets. The idea here is to rule out that

Table 2
Investment returns of venture capitalists with different experiences

	Ln(<i>IRR_Ave</i>)		Ln(<i>ROI_Ave</i>)	
	(1)	(2)	(3)	(4)
<i>Overseas</i>	−0.0955*** (−3.07)	−0.0968** (−2.58)	−0.1963*** (−5.03)	−0.1588*** (−3.05)
Ln(<i>Duration_Ave</i>)	−0.5762*** (−13.58)	−0.5763*** (−12.08)	−0.3375*** (−7.35)	−0.3739*** (−6.88)
<i>Gender</i>		0.0326 (0.69)		−0.0649 (−1.03)
<i>Edu_Master</i>		−0.0479 (−1.07)		−0.1152** (−2.15)
<i>Edu_PhD</i>		−0.0329 (−0.60)		−0.0662 (−0.92)
<i>Experience</i>		−0.0024** (−2.41)		−0.0050*** (−4.01)
<i>VC_Mainland</i>		0.0245 (0.67)		0.0561 (1.07)
<i>Comp_Age</i>		−0.0016*** (−4.39)		−0.0027*** (−5.87)
<i>Industry</i>		0.0083 (0.20)		0.0215 (0.35)
<i>City</i>		−0.0645 (−1.44)		−0.1545** (−2.45)
<i>Stage1</i>		−0.3056 (−1.32)		−0.7601*** (−2.86)
<i>Stage2</i>		0.1818 (0.93)		0.1483 (0.74)
<i>Stage3</i>		0.1510 (0.75)		−0.0070 (−0.04)
Constant	1.3468*** (19.34)	1.4051*** (7.02)	1.2653*** (17.77)	1.7550*** (8.09)
Observations	948	720	947	715
<i>R</i> ²	0.208	0.241	0.069	0.182

This table reports the ordinary least squares regression results for the investment returns of venture capitalists, with *IRR_Ave*, *ROI_Ave*, and *Duration* winsorised to limit the influence of outliers. The dependent variable is the natural logarithm of 1 plus the venture capitalist's weighted average IRR or weighted average ROI generated from 2001 to 2012. The main variables of interest are the venture capitalist's foreign experience dummy (*Overseas*); the time horizon of investment (*Ln(Duration_Ave)*); the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry*, *City* and *Stage*. The unit of this table is a venture capitalist. We construct the series of personal level variables based on deal-level data. For example, *Comp_Age* is the average age of entrepreneurial firms backed by the same venture capitalist. Variable definitions are provided in Appendix Table A1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

Table 3
IPO rate of venture capitalists with different experiences

	<i>Ln(IPOrate)</i>			
	(1)	(2)	(3)	(4)
<i>Overseas</i>	−0.0413*** (−3.42)	−0.1084*** (−6.98)	−0.0537*** (−2.99)	−0.0487** (−2.46)
<i>Ln(Duration_Ave)</i>		−0.0458** (−2.57)		−0.0057 (−0.28)
<i>Gender</i>			−0.0461** (−2.00)	−0.0407* (−1.68)
<i>Edu_Master</i>			−0.0200 (−1.02)	−0.0270 (−1.32)
<i>Edu_PhD</i>			−0.0128 (−0.50)	−0.0039 (−0.14)
<i>Experience</i>			−0.0010* (−1.89)	−0.0037*** (−7.77)
<i>VC_Mainland</i>			0.0499*** (2.78)	0.0224 (1.12)
<i>Comp_Age</i>			−0.0000 (−0.32)	−0.0002 (−1.20)
<i>Industry</i>			−0.0184 (−0.90)	−0.0636*** (−2.74)
<i>City</i>			−0.0449** (−2.09)	−0.0636*** (−2.68)
<i>Stage1</i>			−0.3797*** (−5.23)	−0.5383*** (−5.57)
<i>Stage2</i>			−0.0900 (−1.52)	−0.1542** (−2.16)
<i>Stage3</i>			−0.0895 (−1.50)	−0.2360*** (−3.31)
Constant	0.2583*** (31.42)	0.5380*** (19.50)	0.4974*** (7.50)	0.8109*** (10.33)
Observations	2,133	974	1,295	735
<i>R</i> ²	0.005	0.050	0.097	0.241

This table reports the ordinary least squares regression results for the IPO rates of venture capitalists. The dependent variable is the natural logarithm of 1 plus the IPO rate of venture capitalists generated between 2001 and 2012 (*Ln(IPOrate)*). The main variables of interest are the venture capitalist's foreign experience dummy (*Overseas*); the time horizon of investment (*Ln(Duration_Ave)*); the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry*, *City*, and *Stage*. The unit of this table is a venture capitalist. We constructed the series of personal level variables based on deal-level data. For example, *Comp_Age* is the average age of entrepreneurial firms backed by the same venture capitalist. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

the venture capitalists with foreign experience are simply proxies for some omitted VC firm attributes. In other words, some VC firms' organisational capital (i.e., firm policies, limited partners' resources, or other attributes) could result in a performance discrepancy because a venture capitalist with domestic experience is more likely to be from domestic-funding VC firms. Moreover, foreign-funding VC firms employ venture capitalists with foreign experience more easily.

Nevertheless, we still find that venture capitalists with foreign experience yield worse investment performances with the subsample shown in Table 4. Additionally, the economic magnitude is larger: all else being equal, venture capitalists with foreign experience achieve 13.40, 15.95, and 7.03 percent (or 10.09 percent when considering *Duration*) less average *IRR*, *ROI*, and *IPO* rate, respectively. For brevity, we only present regression results using all control variables. The aforementioned findings are thus consistent with Hypothesis 1.

4.2. Deal-level analysis

In this subsection, we estimate the effect of venture capitalist's foreign experience on investment performance with the deal-level sample. The model we use is similar to Model (1); every venture capitalist-level variable is replaced with a relevant deal-level variable. Notably, we use *ROUND* instead of *Stage* to control for the start-up's status more precisely and *YEAR* to control for any changes in market rules or economy-wide shocks. As before, we focus not only on the deal-level rate of return data but also on whether the capital has an *IPO* exit.

As per columns (1)–(4) of Table 5, venture capitalists with foreign experience are associated with significantly lower returns. Specifically, as reported in columns (2) and (4), a deal invested by a venture capitalist with foreign experience received 6.68 percent lower *IRR* and 11.48 percent lower *ROI* than a deal invested by those with domestic experience.

In columns (5) and (6), we examine the investment's *IPO*-exit possibility, considering that a large number of deals do not file an exit and return in our sample. Specifically, we do not control $\ln(\textit{Duration})$ to prevent eliminating any deals that have not exited from the sample. After controlling for a comprehensive set of possible determinants for *IPO* consequences, we observe that the sample quantity increases to 6,541 in column (6). The coefficient estimates of the *Overseas* dummy are all negative and significant at the 1 percent level, regardless of whether the control variables are controlled. Therefore, Hypothesis 1 is verified to be valid at the deal level.

4.3. Propensity score matching analysis

A reasonable concern regarding the analysis thus far, is that our study only focuses on the sample of venture capitalists whose bios are filed in the database.

Table 4
Investment performance of venture capitalists with different experiences (subsample)

	<i>Ln(IRR_Ave)</i>	<i>Ln(ROI_Ave)</i>	<i>Ln(IPOrate)</i>	
	(1)	(2)	(3)	(4)
<i>Overseas</i>	−0.1340*** (−2.64)	−0.1595** (−2.10)	−0.1009*** (−3.44)	−0.0703** (−2.30)
<i>Ln(Duration_Ave)</i>	−0.5927*** (−9.70)	−0.2910*** (−3.90)		−0.0061 (−0.21)
<i>Gender</i>	−0.0145 (−0.22)	−0.1225 (−1.41)	−0.0506 (−1.31)	−0.0340 (−0.96)
<i>Edu_Master</i>	0.0378 (0.79)	−0.0433 (−0.61)	−0.0369 (−1.21)	−0.0406 (−1.41)
<i>Edu_PhD</i>	0.0658 (0.89)	−0.0256 (−0.25)	−0.0208 (−0.51)	0.0167 (0.41)
<i>Experience</i>	−0.0064*** (−3.90)	−0.0098*** (−3.73)	−0.0034*** (−2.72)	−0.0071*** (−6.58)
<i>Comp_Age</i>	−0.0011*** (−2.98)	−0.0022*** (−3.78)	0.0001 (0.50)	−0.0002 (−1.00)
<i>Industry</i>	0.0011 (0.02)	0.0076 (0.10)	−0.0168 (−0.52)	−0.0573* (−1.80)
<i>City</i>	−0.1113** (−2.17)	−0.2455*** (−3.25)	−0.0424 (−1.37)	−0.0397 (−1.31)
<i>Stage1</i>	2.5266 (1.59)	−0.6108 (−0.14)	−0.0448 (−0.25)	−1.7586 (−0.98)
<i>Stage2</i>	2.8911* (1.85)	0.5161 (0.12)	0.3605** (2.20)	−1.0611 (−0.60)
<i>Stage3</i>	2.6915* (1.72)	0.1947 (0.04)	0.2928* (1.78)	−1.1802 (−0.66)
Constant	−1.2201 (−0.78)	1.4157 (0.33)	0.1485 (0.87)	1.7706 (0.99)
Observations	340	334	555	348
<i>R</i> ²	0.348	0.228	0.153	0.302

This table reports the ordinary least squares regression results for the investment returns and IPO rates of venture capitalists after excluding those who worked for foreign-owned or mixed venture capital firms. The dependent variable are *Ln(IRR_Ave)*, *Ln(ROI_Ave)*, and *Ln(IPOrate)*. Their definitions are the same as in Tables 2 and 3. The main variables of interest are the venture capitalist's foreign experience dummy (*Overseas*); the time horizon of investment (*Ln(Duration_Ave)*); the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry*, *City*, and *Stage*. The unit of this table is the venture capitalist. We constructed the series of personal level variables based on deal-level data. For example, *Comp_Age* is the average age of entrepreneurial firms backed by the same venture capitalist. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

Table 5

Investment performance of venture capitalists with different experiences (deal level)

	<i>Ln (IRR)</i>		<i>Ln (ROI)</i>		<i>IPO</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Overseas</i>	−0.0750*** (−4.28)	−0.0668*** (−2.74)	−0.1460*** (−5.54)	−0.1148*** (−3.21)	−0.2354*** (−8.09)	−0.2551*** (−5.80)
<i>Ln(Duration)</i>	−0.5354*** (−26.55)	−0.5473*** (−18.34)	−0.3209*** (−10.77)	−0.3508*** (−8.77)		
<i>Gender</i>		−0.0234 (−0.70)		−0.0564 (−1.09)		−0.0342 (−0.55)
<i>Edu_Master</i>		−0.0213 (−0.84)		−0.0680* (−1.75)		−0.0592 (−1.16)
<i>Edu_PhD</i>		0.0099 (0.29)		−0.0293 (−0.57)		−0.0540 (−0.83)
<i>Experience</i>		0.0011 (1.39)		0.0018** (1.96)		−0.0005 (−0.62)
<i>VC_Mainland</i>		0.0195 (0.78)		0.0676* (1.83)		0.2569*** (5.57)
<i>Comp_Age</i>		−0.0067*** (−4.22)		−0.0129*** (−6.00)		0.0146*** (5.91)
<i>City</i>		−0.0086 (−0.36)		−0.0235 (−0.67)		0.0380 (0.83)
<i>Industry</i>		0.0042 (0.19)		0.0153 (0.46)		−0.1570*** (−3.92)
Round		yes		yes		yes
Year	yes	yes	yes	yes	yes	yes
Constant	1.3102*** (18.63)	2.0480*** (5.32)	1.4155*** (13.45)	2.0662*** (11.31)	−0.0691 (−0.57)	0.1711 (0.32)
Observations	2,105	1,616	2,106	1,610	10,253	6,541
<i>R</i> ²	0.384	0.394	0.222	0.245		

This table reports the results for investment performance of venture capitalists. The unit of this table is venture capital deal. In the ordinary least squares regression (columns (1)–(4)), the dependent variable is the natural logarithm of 1 plus the IRR or ROI of deals from 2001 to 2012. The main variables of interest are the venture capitalist's foreign experience dummy (*Overseas*); the time horizon of investment (*Ln(Duration_Ave)*); the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and firm's *Comp_Age*, *Industry* and *City*. In addition, we include sequence and year fixed effects at the deal level. Variable definitions are provided in Appendix 1. In the probit regression (columns (5)–(6)), the dependent variable *IPO* is a dummy equals 1 if the deal-related entrepreneurial firm obtains IPO before 2017. Specifically, we do not control *Ln(Duration)* to prevent eliminating any deals that have not exited from the sample. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

Our results may be driven by the fact that some information of the venture capitalists with foreign experience and their outstanding performance has never been captured for some reason (e.g., those who have been abroad for a long

time). Moreover, some venture capitalists with domestic experience may avoid disclosing their personal experiences to the media because of a poor investment performance or an underwhelming educational background. If our analysis is based on the sample, which introduces the survivorship bias issue, then drawing conclusions solely based on the above analysis could be misleading.

To describe the difference between the deals held by the two venture capitalist groups in the observable characteristics, we report the univariate comparisons between these two kinds of deal in columns (1)–(3) of Panel A of Table 6. The results suggest that venture capitalists with foreign experience are more gender-balanced, have a higher level of education and experience, and hold more funds from overseas. Additionally, the entrepreneurial firms backed by them are younger and come from more TMT industries and VC-concentrated cities. Given that the characteristics of these two groups of deal differ, we compare the investment performance within the propensity score matching (PSM) pairs of VC deal in this section.

Subsequently, we use the propensity score from the ‘Prematch’ probit regression (the results are in the column (7) of Panel B) and perform a nearest-neighbour PSM with replacement. As the number of deals held by the venture capitalists with domestic experience exceeds the number of deals held by their counterparts, we use the one, three, and five nearest neighbouring deals of the venture capitalists with domestic experience as our main control group deals.

We conduct a diagnostic test to assess the accuracy of the matching procedure of 1:1. First, we report the univariate comparison between the matched pairs and the results in columns (4)–(6) of Panel A; we observe statistically insignificant differences between the two groups across all but two characteristics (*Gender* and *Edu_PhD*). Accordingly, in unreported analysis, we do not find that there are variables have a bias bigger than 10 percent except for *Gender* (the bias is 12.1 percent) in the ‘Postmatch’ sample. The matching result is acceptable as a whole. Next, we rerun the probit model restricted to the matched sample and report the results in column (8) of Panel B, labelled ‘Postmatch.’ The magnitude of the probit regression coefficients and pseudo R^2 decline dramatically.

Finally, Figures 1 and 2 show the kernel density functions that more directly indicate no significant differences between the treatment group and the control group after using PSM. This aspect suggests that the matching process has removed meaningful differences at the deal level between the two groups.

Table 7 reports the investment return analysis using the propensity score-matched pairs of deal. We report a wide set of results of the different numbers of nearest neighbours used. Panel A reports the *IRR*, *ROI*, and *IPO* differences between the two groups. Even after non-parametrically controlling for other characteristics (PSM), venture capitalists with foreign experience have lower investment returns. Specifically, they yield 5–10 percent less *IRR* and 11–18 percent less *ROI*. We report the results of regression in Panel B. Additionally,

Table 6
Propensity score matching: Diagnostic tests

	Panel A: comparing sample characteristics					Panel B: probit regressions		
	Prematch		Postmatch			Prematch		Postmatch
	<i>Overseas</i> = 0 (1)	<i>Overseas</i> = 1 (2)	<i>Difference</i> (3)	<i>Overseas</i> = 0 (4)	<i>Overseas</i> = 1 (5)	<i>Difference</i> (6)	(7)	(8)
<i>Ln(Duration)</i>	1.4231	1.3552	0.0679*** (3.45)	1.4112	1.4002	0.0110 (0.51)	−0.1573* (−1.83)	−0.1498* (−1.80)
<i>Gender</i>	0.926	0.9062	0.0198*** (3.54)	0.9458	0.9106	0.0353*** (2.73)	−0.1258 (−1.01)	−0.3113** (−2.44)
<i>Edu_Master</i>	0.6951	0.7009	−0.0058 (−0.55)	0.6801	0.6927	−0.0126 (−0.54)	0.0569 (0.55)	−0.1322 (−1.38)
<i>Edu_PhD</i>	0.1404	0.1631	−0.0228*** (−2.75)	0.1889	0.1549	0.0340* (1.80)	0.2932** (2.23)	−0.2412** (−2.03)
<i>Experience</i>	11.7940	18.7165	−6.9226*** (−12.31)	10.3098	11.5302	−1.2204 (−1.17)	0.0054*** (2.80)	0.0037*** (2.04)
<i>VC_Mainland</i>	0.6062	0.1566	0.4496*** (52.17)	0.1700	0.1662	0.0038 (0.20)	−1.4829*** (−19.46)	0.0053 (0.06)
<i>Comp_Age</i>	7.1162	5.3515	1.7646***	7.0391	6.6915	0.3476	−0.0032	−0.0059

(continued)

Table 6 (continued)

	Panel A: comparing sample characteristics				Panel B: probit regressions			
	Prematch		Postmatch		Prematch		Postmatch	
	<i>Overseas</i> = 0 (1)	<i>Overseas</i> = 1 (2)	<i>Difference</i> (3)	<i>Overseas</i> = 0 (4)	<i>Overseas</i> = 1 (5)	<i>Difference</i> (6)	<i>Prematch</i> (7)	<i>Postmatch</i> (8)
<i>City</i>	0.7192	0.8217	(11.22) −0.1025***	0.7746	0.7935	(1.14) −0.0189	(−0.53) 0.1695*	(−1.00) 0.1107
<i>Industry</i>	0.4300	0.5595	(−12.26) −0.1296***	0.4748	0.4773	(0.91) −0.0025	(1.94) 0.1707**	(1.30) −0.0562
<i>Round</i>	no	no	(−13.06) no	no	no	(−0.10) no	(2.10) yes	(−0.75) yes
<i>Year</i>	no	no	no	no	no	no	yes	yes
<i>Observations</i>	4305	5948	10253	794	794	1,588	1,646	1,588
<i>Pseudo R²</i>							0.290	0.022

This table presents the diagnostic tests of the propensity score matching. Panel A reports the pairwise comparisons of variables (except for *Industry* and *Year* indicator variables) on which the matching is performed both prematch and postmatch. Panel B reports the parameter estimates from the probit model used in estimating the propensity scores for the treatment and control groups. The dependent variable in Panel B equals 1 if the deal is financed by a venture capitalist with foreign experience (treatment deal) and 0 otherwise (control deal). The probit is run at the deal level. The Prematch column contains the parameter estimates of the probit estimated on the entire sample without missing values. This model is used to generate the propensity scores for matching. The Postmatch column contains the parameter estimates of the probit estimated on the subsample of matched treatment and control observations. Variable definitions are provided in Appendix 1. The *t*-statistics for the comparison of mean tests are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

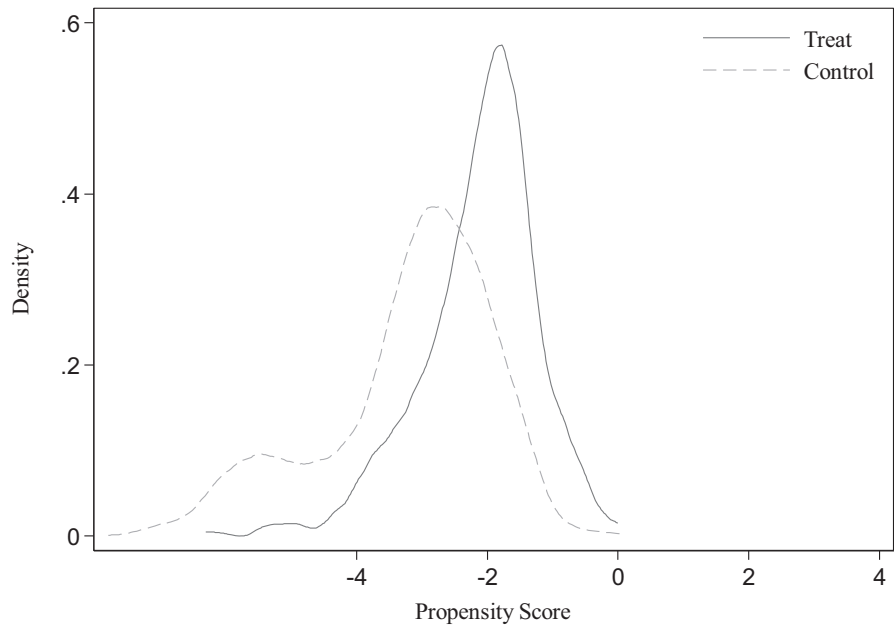


Figure 1 Kernel density of the treat and control groups before matching.

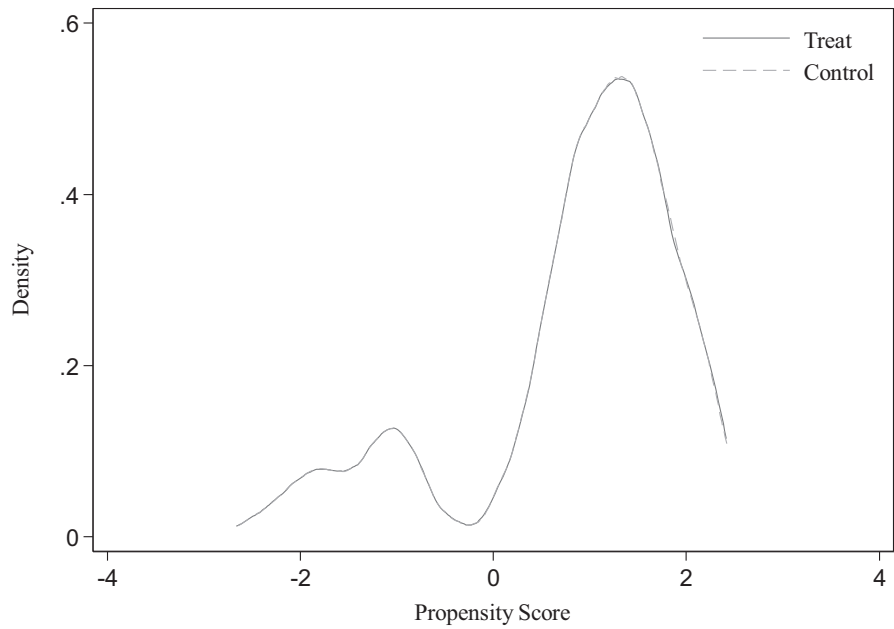


Figure 2 Kernel density of the treat and control groups after matching.

Table 7
Propensity score matching results

	<i>Ln(IRR)</i>			<i>Ln(ROI)</i>			<i>IPO</i>		
	K = 1 (1)	K = 3 (2)	K = 5 (3)	K = 1 (4)	K = 3 (5)	K = 5 (6)	K = 1 (7)	K = 3 (8)	K = 5 (9)
Panel A: comparison									
<i>Overseas</i> = 1	0.5167	0.5167	0.5167	0.7391	0.7391	0.7391	0.8174	0.8174	0.8174
<i>Overseas</i> = 0	0.6214	0.5778	0.5716	0.9279	0.8614	0.8527	0.8753	0.8463	0.8413
Difference	-0.1047***	-0.0612***	-0.0549***	-0.1889***	-0.1223***	-0.1136***	-0.0579***	-0.0290*	-0.0239*
<i>T-Value</i>	(-4.31)	(-3.04)	(-2.93)	(-5.79)	(-4.43)	(-4.41)	(-3.21)	(-1.92)	(-1.67)
Observations	1588	3176	4764	1567	3138	4697	1588	3176	4764
Panel B: regression									
<i>Overseas</i>	-0.1112***	-0.0784***	-0.0821***	-0.1915***	-0.1574***	-0.1597***	-0.3650***	-0.2818***	-0.2933***
	(-6.11)	(-4.78)	(-5.23)	(-6.95)	(-6.66)	(-7.04)	(-4.41)	(-4.01)	(-4.11)
<i>Ln(Duration)</i>	-0.5718***	-0.4939***	-0.4840***	-0.3240***	-0.2337***	-0.2286***	-0.1965*	-0.3398***	-0.3572***
	(-19.04)	(-22.99)	(-28.10)	(-8.12)	(-8.02)	(-9.69)	(-1.68)	(-3.89)	(-4.75)
<i>Gender</i>	-0.0363	-0.1481***	-0.1434***	-0.0959	-0.2028***	-0.2036***	0.2341	-0.1027	-0.1313
	(-0.95)	(-4.78)	(-5.92)	(-1.64)	(-4.59)	(-5.87)	(1.51)	(-0.85)	(-1.30)
<i>Edu_Master</i>	-0.0169	-0.0446**	-0.0153	-0.0997**	-0.1246***	-0.0907***	0.0281	0.1451	0.2637***
	(-0.66)	(-2.54)	(-1.05)	(-2.49)	(-4.44)	(-3.91)	(0.22)	(1.51)	(3.12)
<i>Edu_PhD</i>	-0.0198	-0.0119	-0.0181	-0.1109**	-0.0729**	-0.1009***	0.2112	0.0905	0.0367
	(-0.64)	(-0.58)	(-1.06)	(-2.24)	(-2.13)	(-3.63)	(1.30)	(0.80)	(0.38)
<i>Experience</i>	0.0007	0.0008**	0.0015***	0.0008	0.0004	0.0013**	0.0024	0.0050***	0.0051***
	(1.40)	(2.02)	(4.10)	(0.72)	(0.59)	(2.28)	(1.06)	(2.89)	(3.42)
<i>VC_Mainland</i>	0.0090	0.0088	0.0186	0.0314	0.0474*	0.0521**	-0.0805	-0.1821**	-0.1989***
	(0.33)	(0.47)	(1.27)	(0.78)	(1.73)	(2.40)	(-0.72)	(-2.32)	(-2.97)
<i>Comp_Age</i>	-0.0082***	-0.0102***	-0.0071***	-0.0140***	-0.0147***	-0.0104***	-0.0311***	-0.0178***	-0.0238***
	(-4.69)	(-7.66)	(-7.17)	(-5.64)	(-7.93)	(-7.43)	(-4.16)	(-3.32)	(-5.62)
<i>City</i>	-0.0177	-0.0568***	-0.0667***	-0.0386	-0.0958***	-0.1170***	0.1902*	0.2377***	0.2498***

(continued)

Table 7 (continued)

	<i>Ln(IRR)</i>			<i>Ln(ROI)</i>			<i>IPO</i>		
	K = 1 (1)	K = 3 (2)	K = 5 (3)	K = 1 (4)	K = 3 (5)	K = 5 (6)	K = 1 (7)	K = 3 (8)	K = 5 (9)
<i>Industry</i>	(-0.70) 0.0500** (2.35)	(-3.24) -0.0019 (-0.12)	(-4.77) 0.0187 (1.47)	(-1.10) 0.1212*** (3.51)	(-3.86) 0.0792*** (3.24)	(-5.87) 0.1029*** (5.21)	(1.76) -0.1755* (-1.81)	(2.97) -0.0210 (-0.30)	(3.74) 0.0221 (0.36)
<i>Round</i>	yes	yes	yes	yes	yes	yes	yes	yes	yes
<i>Year</i>	yes	yes	yes	yes	yes	yes	yes	yes	yes
<i>Constant</i>	1.5510*** (16.63)	2.1944*** (4.45)	2.1794*** (4.35)	1.5661*** (10.74)	1.4403*** (12.44)	1.4438*** (13.80)	4.5394*** (13.44)	-0.6237 (-0.81)	-0.5234 (-0.69)
<i>Observations</i>	1,588	3,176	4,764	1,567	3,138	4,697	1,588	3,176	4,764
<i>R</i> ²	0.461	0.414	0.415	0.311	0.305	0.310	0.177	0.215	0.253

This table reports the results of Propensity Score Matching. Panel A presents the differences in mean natural logarithm of IRR, ROI and IPO rate, based on two groups of deals financed by venture capitalists with different experience, using the propensity score matching algorithm. The treatment group is defined as deal financed by venture capitalist with foreign experience, and the control group is defined as deal financed by venture capitalist without foreign experience. *Difference* is the average difference in the *Ln(IRR)*, *Ln(ROI)* or *IPO* between treated and matched control deals. Panel B reports the results of the OLS regression and probit regression. The dependent variable is *Ln(IRR)*, *Ln(ROI)* or *IPO*. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * indicate significance at 1, 5, and 10 percent levels, respectively.

we examine the IPO rates between the two types of deals using matched sample and obtain similar results. Therefore, Hypothesis 1 is still valid.

4.4. Different foreign experience types

Further, we explore whether different foreign experiences have disparate effects on investment performance. Compared to education experience, work experience provides better access to local business for individuals and benefits their human capital (Bauernschuster *et al.*, 2009). Therefore, we expect the relationship between venture capitalists' foreign experience and their performance to be stronger for those with foreign working experience.

To test this hypothesis, we run the regressions; Table 8 shows the results. The dummies *Overseas_work* and *Overseas_study* are proxies for venture capitalists' foreign work and education experiences, respectively. These are used as the independent variables of regressions on investment performance because some venture capitalists have both foreign work and study experiences.

Table 8 shows that the results are similar and consistent across all three investment performance proxies. In all cases, the coefficient for *Overseas_work* is negative and significant but that for *Overseas_study* is not. Additionally, the former is quantitatively greater than the latter.

4.5. Investment strategies

To test Hypothesis 3, we investigate the investment strategies of venture capitalists with different experiences. As per our analysis, if venture capitalists stay in a strong bargaining position, they tend to adopt aggressive investment strategies to maximise the return. Conversely, they take prudent or cooperative strategies to avoid risk. Table 9 reports the results of a probit analysis that uses deal-level data. The independent variable is the dummy *Overseas*, which is denoted by 1 for deals conducted by venture capitalists with foreign experience and 0 for those with domestic experience. The dependent variables are *Multistage*, a dummy that equals 1 for deals that are executed with multistage capital injection and 0 for one-time deals; *Syndicate*, a dummy that equals 1 for deals executed by more than 1 investor and 0 for sole investor deals; and *Leader*, a dummy that equals 1 if the venture capitalist of the deal invests earlier than other venture capitalists in the same round and 0 otherwise.

Venture capitalists with foreign experience show prudent investment strategies compared to their counterparts. Specifically, they tend to invest with multistage and syndicate strategies but always give up being leaders. As per Table 9, the coefficient estimates of *Multistage* and *Syndication* are positive and significant at the 1 percent level, except for column (4), and the coefficient estimate of *Leader* is significantly negative.

Further, we probe into venture capitalists' supervision and management for the portfolio firms after investing. Generally, directorship is the primary way to

Table 8
Types of foreign experience and investing performance

	<i>Ln(IRR)</i>		<i>Ln(ROI)</i>		<i>IPO</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Overseas_work</i>	−0.0468* (−1.89)		−0.1141*** (−3.04)		−0.1693*** (−3.71)	
<i>Overseas_study</i>		−0.0456* (−1.73)		−0.0631 (−1.64)		−0.1432*** (−3.26)
<i>Ln(Duration)</i>	−0.5424*** (−17.39)	−0.5238*** (−15.70)	−0.3346*** (−7.82)	−0.3399*** (−7.79)		
<i>Gender</i>	0.0053 (0.16)	−0.0216 (−0.55)	−0.0128 (−0.24)	−0.0558 (−0.94)	−0.0044 (−0.07)	−0.0320 (−0.48)
<i>Edu_Master</i>	−0.0185 (−0.68)	−0.0174 (−0.54)	−0.0682 (−1.64)	−0.0394 (−0.86)	−0.0473 (−0.88)	0.0069 (0.12)
<i>Edu_PhD</i>	0.0116 (0.32)	0.0265 (0.65)	−0.0235 (−0.43)	0.0124 (0.21)	−0.0162 (−0.23)	0.0144 (0.20)
<i>Experience</i>	0.0000 (0.04)	0.0012 (1.38)	0.0002 (0.29)	0.0019* (1.95)	−0.0024** (−2.00)	−0.0004 (−0.57)
<i>VC_Mainland</i>	0.0234 (0.93)	0.0188 (0.67)	0.0502 (1.32)	0.0677 (1.63)	0.2889*** (5.89)	0.2122*** (4.40)
<i>Comp_Age</i>	−0.0081*** (−5.28)	−0.0081*** (−4.50)	−0.0138*** (−6.42)	−0.0151*** (−6.51)	0.0146*** (5.62)	0.0197*** (6.30)
<i>City</i>	−0.0020 (−0.08)	0.0135 (0.45)	−0.0236 (−0.64)	0.0160 (0.39)	0.0566 (1.14)	0.0673 (1.33)
<i>Industry</i>	−0.0021 (−0.09)	−0.0077 (−0.31)	0.0210 (0.59)	−0.0037 (−0.10)	−0.1512*** (−3.50)	−0.1363*** (−3.19)
Round	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes
Constant	1.9795*** (5.19)	1.9131*** (5.04)	1.9325*** (8.97)	1.7521*** (9.59)	−0.2231 (−0.40)	−0.0941 (−0.18)
Observations	1,399	1,341	1,392	1,334	5,636	5,959
<i>R</i> ²	0.408	0.380	0.257	0.240		

This table reports the results for the investment performance of venture capitalists and their foreign experience while working or studying. The unit of this table is the venture capital deal. In the ordinary least squares regression (columns (1)–(4)), the dependent variable is the natural logarithm of 1 plus the *IRR* or *ROI* of deals from 2001 to 2012. The main variables of interest are the venture capitalist's foreign experience dummy *Overseas_work* and *Overseas_study*; the time horizon of investment *Ln(Duration)*; the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry* and *City*. In addition, we include sequence and year fixed effects at the deal level. Variable definitions are provided in Appendix 1. In the probit regression (columns (5)–(6)), the dependent variable *IPO* is a dummy equals 1 if the deal-related entrepreneurial firm obtained IPO before 2017. Specifically, we do not control *Ln(Duration)* in columns (5)–(6) to prevent eliminating any deals that have not exited from the sample. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

monitor and provide services to entrepreneurial firms for venture capitalists. However, venture capitalists with foreign experience may forgo being directors of portfolio companies and select other risk-control strategies to avoid conflicts with management.

Table 10 empirically studies whether the venture capitalists with different experiences show disparate director participation rates. We define a *Director* dummy that equals 1 if the venture capitalist appeared in the director list of the

Table 9
Investment strategies of venture capitalists

	<i>Multistage</i>		<i>Syndicate</i>		<i>Leader</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Overseas</i>	0.2659*** (9.82)	0.4078*** (4.64)	0.1507*** (5.80)	0.1583* (1.93)	−0.1579*** (−5.24)	−0.2540*** (−2.93)
<i>Ln(Duration)</i>		0.5996*** (6.77)		−0.1423* (−1.76)		0.0847 (1.02)
<i>Gender</i>		0.0271 (0.22)		−0.0863 (−0.74)		−0.1257 (−1.02)
<i>Edu_Master</i>		−0.0339 (−0.35)		0.0547 (0.61)		0.0468 (0.50)
<i>Edu_PhD</i>		−0.1081 (−0.83)		0.0854 (0.72)		0.1186 (0.95)
<i>Experience</i>		0.0033** (2.04)		−0.0025* (−1.67)		0.0022 (1.26)
<i>VC_Mainland</i>		−0.0299 (−0.34)		−0.0214 (−0.26)		−0.0272 (−0.31)
<i>Syndicate</i>		−0.1112 (−1.49)				
<i>Multistage</i>				−0.1045 (−1.35)		0.0430 (0.53)
<i>Comp_Age</i>		0.0051 (0.84)		−0.0116** (−2.04)		0.0212** (2.49)
<i>City</i>		0.1197 (1.37)		0.0849 (1.07)		−0.0263 (−0.31)
<i>Industry</i>		0.2655*** (3.38)		0.2284*** (3.05)		−0.1203 (−1.56)
Round		yes		yes		Yes
Year	yes	yes	yes	yes	yes	Yes
Constant	−0.9080*** (−6.76)	−1.8985** (−2.20)	−0.3183** (−2.57)	4.0899*** (11.80)	1.3885*** (8.11)	1.3123 (1.45)
Observations	10,253	1,624	10,040	1,624	9,421	1,633

This table reports the results for investment strategies of venture capitalists with different experiences. The unit of this table is the venture capital deal. In the probit model, the dependent variables are dummies: *Multistage*, *Syndicate*, and *Leader*. The main variables of interest are the venture capitalist's foreign experience dummy *Overseas*; the time horizon of investment *Ln(Duration)*; the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry* and *City*. In addition, we include sequence and year fixed effects at the deal level. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

portfolio firm and 0 otherwise. Based on the coefficient estimate of *Overseas* dummy, venture capitalists with foreign experience are less likely to acquire a seat on the portfolio firm's board than those with domestic experience.

Table 10

Investment duration and position on the board of venture capitalists

	<i>Ln(Duration)</i>		<i>Director</i>	
	(1)	(2)	(3)	(4)
<i>Overseas</i>	−0.0575*** (−2.98)	−0.0461* (−1.78)	−0.6410*** (−8.07)	−0.4048*** (−3.60)
<i>Ln(Duration)</i>				−0.1013 (−0.78)
<i>Gender</i>		0.0117 (0.32)		−0.1955 (−1.20)
<i>Edu_Master</i>		−0.0014 (−0.05)		−0.0252 (−0.20)
<i>Edu_PhD</i>		0.0309 (0.81)		−0.1449 (−0.85)
<i>Experience</i>		0.0012** (2.41)		−0.0505*** (−6.14)
<i>VC_Mainland</i>		0.0351 (1.33)		−0.0664 (−0.60)
<i>Comp_Age</i>		−0.0000 (−0.02)		−0.0109 (−1.40)
<i>City</i>		0.0262 (1.03)		−0.1481 (−1.38)
<i>Industry</i>		0.0279 (1.18)		−0.0379 (−0.35)
Round		yes		Yes
YearR	yes	yes	yes	Yes
Constant	1.8643*** (28.14)	1.8507*** (7.12)	0.5871** (2.56)	1.2759 (0.96)
Observations	2,149	1,646	1,372	854
<i>R</i> ²	0.089	0.144		

This table reports the results for investment duration and position on the board of venture capitalists with different experience. The unit of this table is the venture capital deal. In the ordinary least squares regression, the dependent variable is *Ln(Duration)*. In the probit model, the dependent variable is a dummy: *Director*. The main variables of interest are the venture capitalist's foreign experience dummy *Overseas*; the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*; and the firm's *Comp_Age*, *Industry* and *City*. In addition, we include sequence and year fixed effects at the deal level. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

We are aware that the fewer directorships of venture capitalists with foreign experience may come from shorter investment durations. In other words, the investment quit before the venture capitalists could acquire a seat on the board. In columns (1) and (2), we find that the investment duration of venture capitalists with foreign experience is 4.61–5.75 percent shorter than that of venture capitalists with domestic experience. Nevertheless, we have not

observed a significant variation of coefficient estimates of *Overseas* after controlling the *Duration* according to columns (3) and (4). Thus, the fewer directorships must stem from other reasons, such as having less control over the company or cultural disparity.

Another reasonable concern regarding this analysis is that the sample only focuses on the related entrepreneurial firms that eventually go public (private firms do not publish the list of directors). However, at least partially, the analysis still demonstrates that venture capitalists with foreign experience participate less in the corporation's supervision and management than their counterparts. The above findings are thus consistent with Hypothesis 3.

5. Mechanisms analysis

Our analysis thus far shows that venture capitalists with foreign experience yield lower returns and IPO rates than venture capitalists with domestic experience. The next natural question is regarding the economic mechanisms. In this section, we examine the potential mechanisms from the following two aspects: (1) treatment or selection effect as the direct cause; and (2) cultural distance as the natural reason.

5.1. Treatment and selection effects

We are interested in the mechanism that drives our main effect, in particular, whether venture capitalists with foreign experience perform worse because of their inferior ability to nurture start-ups (a treatment effect) or their inferior ability to screen firms with a high potential or good price (a selection effect). To disentangle the two effects, we employ a Heckman two-stage regression because the selection effect implies a potential endogeneity or selection bias issue in our result. The question of the two effects is ultimately a question about the causal effect, that is, good venture capitalists bring excellent deal performance, or rare potential deals attract 'good' venture capitalists and then yield good performance. In other words, the better performance of venture capitalists with domestic experience is driven by selection bias in that they are merely picking companies of better quality. Hence, we attempt to solve the endogeneity problem through the instrumental variable method. By referring to Bottazzi *et al.* (2008), we construct an instrumental variable for the proportion of venture capitalists with foreign experience in the portfolio firm's city (*IV_Oversea*).

The selected instrumental variable should be appropriate. First, the proportion of venture capitalists with foreign experience of one city has little direct effect on a particular local VC deal's return and exit, and the relevant entrepreneurial firm only contacts its own investors. Second, a strong correlation may exist between the instrumental variable and the independent variable (*Overseas*). According to a previous study, the choices of VC and

entrepreneurial firm exhibit strong local bias, that is, venture capitalists are likely to sign agreements with entrepreneurial firms located in the same area (Fracassi, 2016). Thus, an entrepreneurial firm is likely to cooperate with a venture capitalist that has foreign experience if the city has many venture capitalists with foreign experience.

Table 11 reports the effect of the venture capitalists with foreign experience on investment returns after using the instrument variable. The endogenous

Table 11
Investment returns of venture capitalists: Instrument variable approach

	<i>Ln(IRR)</i> (1)	<i>Overseas</i> (2)	<i>Ln(ROI)</i> (3)	<i>Oversea</i> (4)	<i>IPO</i> (5)	<i>Overseas</i> (6)
<i>Overseas</i>	−0.4986*** (−7.39)		−0.8415*** (−8.46)		−0.7615*** (−22.11)	
<i>IV_Overseas</i>		0.8774*** (4.27)		0.8494*** (4.30)		0.8350*** (9.18)
<i>Ln(Duration)</i>	−0.5603*** (−21.91)	−0.2276*** (−2.70)	−0.3742*** (−9.74)	−0.2178*** (−2.61)		
<i>Gender</i>	−0.0322 (−0.89)	−0.0664 (−0.56)	−0.0761 (−1.39)	−0.0744 (−0.63)	−0.0674*** (−3.11)	−0.2484*** (−4.25)
<i>Edu_Master</i>	−0.0263 (−0.90)	0.0153 (0.15)	−0.0715 (−1.63)	0.0278 (0.27)	−0.0109 (−0.61)	0.0195 (0.39)
<i>Edu_PhD</i>	0.0296 (0.78)	0.1987 (1.53)	0.0138 (0.24)	0.2224* (1.74)	0.0341 (1.51)	0.1964*** (3.11)
<i>Experience</i>	0.0016*** (3.22)	0.0067*** (3.43)	0.0029*** (3.95)	0.0073*** (3.65)	0.0005** (2.37)	0.0033*** (4.34)
<i>VC_Mainland</i>	−0.2053*** (−4.93)	−1.3857*** (−18.29)	−0.3105*** (−5.01)	−1.3821*** (−18.28)	−0.2792*** (−12.36)	−1.3720*** (−35.43)
<i>Comp_Age</i>	−0.0062*** (−3.50)	−0.0046 (−0.77)	−0.0125*** (−4.64)	−0.0051 (−0.87)	0.0024** (2.51)	−0.0076*** (−2.82)
<i>Industry</i>	0.0300 (1.29)	0.0947 (1.21)	0.0536 (1.52)	0.1110 (1.43)	−0.0093 (−0.69)	0.0453 (1.21)
Round	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	Yes
Constant	2.3088*** (8.59)		2.4767*** (5.24)		1.0201*** (6.18)	
Observations	1,703	1,703	1,703	1,703	6,731	6,731

This table reports the first- and second-stage results of a two-stage least squares regression using the proportion of venture capitalists with foreign experience in the start-up’s city as instrument variable (*IV_Overseas*). The unit of this table is the venture capital deal. In the ordinary least squares regression (columns (1), (3), and (5)), the dependent variables are *Ln(IRR)*, *Ln(ROI)* and *IPO*. In the probit regression (columns (2), (4), and (6)), the dependent variable is the dummy variable *Oversea*. Variable definitions are provided in Appendix 1. Specifically, we do not control *Ln(Duration)* in columns (5)–(6) to prevent eliminating any deals that have not exited from the sample. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

variable (*Overseas*) is a binary variable rather than a continuous one; hence, the traditional two-stage estimation method does not apply to the model. Therefore, we refer to Roodman (2011) and employ the conditional mixed-process estimation method combined with an instrumental variable for regression. Columns (2), (4), and (6) indicate that the instrumental variable is highly correlated with the independent variable. Further, columns (1), (3), and (5) indicate that after eliminating endogeneity, the conclusion that investment returns and IPO rates are significantly and negatively correlated with venture capitalists' foreign experience remains valid. Although we cannot completely ensure or rule out the selection effect, such evidence allows us to attribute part of the performance difference to the treatment effect.

5.2. Cultural distance and investment performance

In this subsection, we examine the role of cultural similarities or differences in investment performance. Culture impresses one's mind through education or working in a particular environment and influences an individual's behavioural pattern. If the cultural difference is a vital reason for the performance discrepancy of the two venture capitalist groups, investment performance should vary according to the culture distance between the venture capitalists' education or work location and their portfolio company's location (Hypothesis 2).

We define cultural distance measures between a venture capitalist's culture-received country and the portfolio company's country based on the values of Hofstede's six cultural dimensions. The Hofstede framework is one of the most cited cultural frameworks in international business, marketing, and organisational development (Kirkman *et al.*, 2006). In Appendix Table A3, we present the cultural distance between China and the related countries as measured by Hofstede's four and six cultural dimensions.

Subsequently, we examine whether venture capitalists' investment performance differs with the cultural distance. There's a strong 'collinearity' between venture capitalist's foreign background and culture distance. Venture capitalists with foreign experience (*Overseas* = 1) related culture distance is greater than zero ($CD_4/CD_6 > 0$), and venture capitalists without foreign experience (*Overseas* = 0) related culture distance is equal to zero ($CD_4/CD_6 = 0$). In order to prevent cultural distance affect investment performance as a proxy of foreign background, we focus only on venture capitalists with foreign background to explore the relation between culture distance and firm's IPO possibility in columns (1) and (2). The negative coefficients on *CD_4* and *CD_6* indicate that farther culture distance is associated with lower IPO possibility. IPO outcome partly reflects whether the cooperation between venture capitalist and firm's founder process is smooth or not. On the other hand, the results in columns (3) and (4) show that IPOs help venture capitalist obtain higher IRR and ROI compared with other exit ways. It is worth noting we have not found

Table 12
Culture distance, IPO and investment return

	Panel A			Panel B		
	Completed deals			Completed deals		
	All deals, Overseas = 1					
	<i>IPO</i>			<i>IPO_Native</i>		
	(1)	(2)		(5)	(6)	<i>Ln(ROI)</i> (7)
<i>CD_4</i>	−0.0146*** (−2.84)					
<i>CD_6</i>		−0.0178*** (−3.06)				
<i>IPO</i>			0.2824*** (11.10)			
<i>Overseas</i>				−1.0583*** (−9.76)		
<i>IPO_Native</i>						0.5185*** (14.34)
<i>IPO_Overseas</i>						0.3050*** (6.59)
<i>Ln(Duration)</i>			−0.5466*** (−18.77)	0.8031*** (6.94)	−0.5750*** (−19.58)	−0.3818*** (−9.69)
<i>Gender</i>	0.0352 (0.45)	0.0283 (0.36)	−0.0080 (−0.24)	0.0716 (0.45)	−0.0112 (−0.34)	−0.0364 (−0.69)
<i>Edu_Master</i>	−0.0511 (−0.74)	−0.0537 (−0.78)	−0.0201 (−0.83)	0.1317 (0.97)	−0.0260 (−1.08)	−0.0756** (−2.05)
<i>Edu_PhD</i>	0.0061 (0.07)	0.0028 (0.03)	0.0099 (0.30)	0.5213*** (2.98)	−0.0066 (−0.20)	−0.0492 (−1.02)
<i>Experience</i>	0.0008 (1.00)	0.0009 (1.14)	0.0012 (1.59)	0.0012 (0.59)	0.0013* (1.68)	0.0021** (2.26)
<i>VC_Mainland</i>	0.1688** (2.32)	0.1681** (2.30)	0.0575*** (2.89)	1.3604*** (11.33)	−0.0058 (−0.26)	0.0521 (1.59)

(continued)

Table 12 (continued)

Panel A			Panel B		
All deals, Overseas = 1			Completed deals		
<i>IPO</i>			<i>Ln(IRR)</i>	<i>Ln(ROI)</i>	
(1)	(2)		(3)	(4)	(5)
<i>Comp_Age</i>	0.0165*** (3.29)	0.0170*** (3.36)	−0.0066*** (−4.05)	−0.0127*** (−5.62)	0.0248** (2.28)
<i>City</i>	0.1100 (1.64)	0.1256* (1.87)	−0.0193 (−0.82)	−0.0392 (−1.15)	−0.4118*** (−3.43)
<i>Industry</i>	−0.2143*** (−3.89)	−0.2171*** (−3.93)	0.0102 (0.49)	0.0218 (0.68)	−0.2526** (−2.37)
Round	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes
Constant	−0.0541 (−0.08)	−0.0546 (−0.09)	1.7834*** (4.09)	1.5329*** (8.62)	1.8554*** (4.10)
Observations	4,015	4,006	1,616	1,610	1,616
<i>R</i> ²	0.108	0.110	0.433	0.302	0.474
					0.315

Panel A in this table reports the relation between culture distance (between venture capitalist's experience country and China) and start-up's IPO possibility and the influence of start-up's IPO on investment return compared with other exit methods. Panel B reports the preference on IPO location for venture capitalist with different experience and the influence of that on the investment return. The unit of this table is venture capital deal. The sample in columns (1)–(2) is all deals invested by foreign background venture capitalists, and the sample in columns (3)–(7) is deals that had completed before 2017. In columns (1)–(2), the dependent variable *IPO* is a dummy variable which equals one if the deal related entrepreneurial firm obtains IPO before 2017. In columns (3)–(4), the dependent variable is the natural logarithm of one plus the IRR or ROI of deals. In column (5), the dependent variable is the dummy *IPO_Native*, which means the start-up's IPO location is mainland China. In columns (6)–(7), the dependent variable is the natural logarithm of one plus the IRR or ROI of deals. The main variables of interest are *Overseas*, *CD_4*, *CD_6*, *IPO_Native* and *IPO_Overseas*. Additionally, we control venture capitalist's *Gender*, *Education*, *Experience*, *VC_Mainland*, firm's age *Comp_Age*, *Industry*, *City*. We also include round and year fixed effects at the deal level. Variable definitions are provided in Appendix 1. Robust *t*-statistics are reported in parentheses. ***, **, and * indicate significance at 1, 5, and 10 percent levels, respectively.

any significant direct correlation between culture distance and investment return. Therefore, cultural differences and IPOs may not be the whole reason for the performance difference, and there are some other reasons which need to be further explored.

In Panel B of Table 12, we explore whether venture capitalists without foreign experience prefer to facilitate the native IPOs of portfolio firms which is more profitable compared with overseas IPOs for venture capitalists. Therefore, Hypothesis 2 is verified to some point.

6. Conclusion

In this study, we provide comprehensive evidence related to whether a venture capitalist's foreign experience impacts their investment performance using a venture capitalist- and deal-level sample. Our results reveal that venture capitalists with foreign experience yield significantly worse investment performances than those with domestic experience, as measured by their investment's average returns and the proportion of investments that successfully exited via IPOs. A further analysis indicates that venture capitalists with foreign experience prefer prudent investment strategies and are less ambitious in terms of being directors. Additionally, we find that the worse performance is driven by the treatment effect rather than the selection effect and that cultural disparity is a natural reason for the differing investment performances of the two venture capitalist groups.

VC has been regarded as an important resource by emerging countries. However, the source of its value is still an open question that has been debated as a crucial policy issue in recent years. This study probes into the role of foreign experience in the VC industry, and the outcomes are useful for policy-makers, corporate managers, financial intermediaries, and investors.

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Appendix

Table A1
Variable definitions

Variable	Description
Panel A: venture capitalists-Level	
<i>IRR_Ave</i>	Venture capitalist's weighted average IRR, which is weighted by all deal amounts invested by the venture capitalist from 2001 to 2012
<i>ROI_Ave</i>	Venture capitalist's weighted average ROI, which is weighted by all deal amounts invested by the venture capitalist from 2001 to 2012
<i>Duration_Ave</i>	Venture capitalist's weighted average investing years, which is weighted by all deal amounts invested by the venture capitalist from 2001 to 2012
<i>IPOrate</i>	The fraction of deals that are successfully exited via an IPO in venture capitalist's all deals from 2001 to 2012

(continued)

Table A1. (Continued)

Variable	Description
<i>Oversea</i>	A dummy variable for a venture capitalist that have foreign university-studying or working experience
<i>IV_Oversea</i>	The proportion of venture capitalists with foreign experience in the portfolio firm's city
<i>Gender</i>	A dummy variable for a venture capitalist that is male
<i>Edu_Master</i>	A dummy variable for a venture capitalist that has a master's degree
<i>Edu_PhD</i>	A dummy variable for a venture capitalist that has a PhD degree
<i>Experience</i>	The total number of investment deals taken by the venture capitalist before
<i>VC_Mainland</i>	A dummy variable for a venture capitalist who works in a mainland-funded VC institution
Panel B: deals-Level	
<i>IRR</i>	The internal rate of return of a investment deal, which equals annual return rate on compound interest
<i>ROI</i>	The return on investment of a deal, which equals annual return rate on simple interest
<i>IPO</i>	A dummy variable for a deal with IPO exit
<i>Duration</i>	The years since capital injection to exit
<i>Multistage</i>	A dummy variable for a deal which is fulfilled by staging of capital infusions
<i>Syndicate</i>	A dummy variable for a deal which involves more than one venture capitalist
<i>Leader</i>	A dummy variable for a deal in which the venture capitalist take the lead role
<i>Director</i>	A dummy variable for a deal in which the venture capitalist take a seat on the entrepreneurial firm's board
<i>CD_4/CD_6</i>	The cultural distance between the country where the venture capitalist received education or worked and the country where the portfolio companies located (China), using the four or six widely used original Hofstede's measures
<i>Comp_Age</i>	Years from the enterprise was founded to capital injection.
<i>City</i>	A dummy variable equal to one if the entrepreneurial firm is founded in Beijing, Guangdong, Shanghai, Zhejiang or Jiangsu where venture capital is most concentrated
<i>Industry</i>	A dummy variable equal to one if the entrepreneurial firm belongs to the telecommunications, media, technology (TMT) industry
<i>Stage</i>	Set four dummy variables, for each of which takes the value one, if the deal took place in firm's initial stage, growth stage, development stage, and profit stage respectively
<i>Round FE</i>	Fixed effects for the investment round: Angel, PE-Growth, VC-series A to VC-series E
<i>Year FE</i>	Fixed effects for the investment year: 2001 to 2012

This table defines the variables used in the analysis and Table 1 provides summary statistics.

Table A2

Investment returns of venture capitalists with different experiences (without winsorising)

	Ln (<i>IRR_Ave</i>)		Ln (<i>ROI_Ave</i>)	
	(1)	(2)	(3)	(4)
<i>Overseas</i>	−0.0940* (−1.87)	−0.0775* (−1.85)	−0.2382*** (−5.51)	−0.1980*** (−3.39)
<i>Ln(Duration_Ave)</i>	−0.7877*** (−6.82)	−0.6650*** (−7.53)	−0.3473*** (−6.86)	−0.4010*** (−6.57)
<i>Gender</i>		−0.0017 (−0.03)		−0.0818 (−1.15)
<i>Edu_Master</i>		−0.0256 (−0.51)		−0.0904 (−1.50)
<i>Edu_PhD</i>		−0.0440 (−0.77)		−0.0142 (−0.18)
<i>Experience</i>		−0.0025** (−2.48)		−0.0056*** (−3.95)
<i>VC_Mainland</i>		0.0132 (0.32)		0.0984* (1.66)
<i>Comp_Age</i>		−0.0020*** (−3.84)		−0.0030*** (−5.90)
<i>Industry</i>		−0.0462 (−0.85)		−0.0204 (−0.30)
<i>City</i>		−0.0892 (−1.24)		−0.0976 (−1.40)
<i>Stage1</i>		−0.3410 (−0.79)		−0.4712 (−1.63)
<i>Stage2</i>		−0.0237 (−0.07)		0.1754 (0.82)
<i>Stage3</i>		−0.0643 (−0.18)		0.0085 (0.04)
Constant	1.6719*** (8.96)	1.8304*** (3.81)	1.3167*** (16.78)	1.7784*** (7.61)
Observations	963	729	963	728
<i>R</i> ²	0.179	0.194	0.066	0.157

This table presents the results from Table 2 without winsorising any of the continuous variables. The dependent variable is the natural logarithm of 1 plus the venture capitalist's weighted average *IRR* or *ROI* generated from 2001 to 2012. The main variables of interest are the venture capitalist's foreign experience dummy *Overseas*, the time horizon of investment *Ln(Duration_Ave)*, the venture capitalist's *Gender*, *Education*, *Experience*, and *VC_Mainland*, and the firm's *Comp_Age*, *Industry*, *City*, and *Stage*. The unit of this table is the venture capitalist. We construct the series of personal level variables based on deal-level data. For example, *Comp_Age* is the average age of entrepreneurial firms backed by the same venture capitalist. Variable definitions are provided in Table A1. Robust *t*-statistics are reported in parentheses. ***, **, and * denote significance at 1, 5, and 10 percent levels, respectively.

Table A3

Cultural distance between China and related countries

	<i>CD_4</i>	<i>CD_6</i>
Singapore	7.2629	6.6130
India	7.8780	8.0502
Malaysia	7.5166	10.7968
South Korea	16.1206	10.9928
Germany	18.4713	12.6489
Thailand	12.3390	12.9035
Japan	19.4245	13.2959
Italy	19.4888	13.7300
Spain	18.0312	14.1231
Switzerland	18.0624	14.1525
France	20.0156	14.4950
Belgium	21.6362	15.4669
Great Britain	20.6322	16.8494
Canada	19.0411	17.0249
Netherlands	23.1800	17.4851
Mexico	13.2618	18.4370
Norway	23.1382	18.5252
USA	20.7861	18.7784
Sweden	23.3506	18.8803
New Zealand	21.3161	18.8914
Australia	21.1098	19.5792

According to Hofstede's explanation, the earliest public version of the instrument (VSM 82) covered four dimensions: Power Distance Index (PDI), Individualism Index (IDV), Masculinity Index (MAS), and Uncertainty Avoidance Index (UAI). The sixth dimension in the present version adds two dimensions: Long Term Orientation Index (LTO) and Indulgence versus Restraint Index (IVR). We use the first four dimensions measure to calculate *CD_4*, and all six dimension measures to calculate *CD_6* in our analysis.